Generative artificial intelligence

Author: Ignasi Sebastià Oriol

Coordination: Montse Guitert, Teresa Romeu and Marc Romero

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What is generative artificial intelligence and how can it benefit us?

Introduction

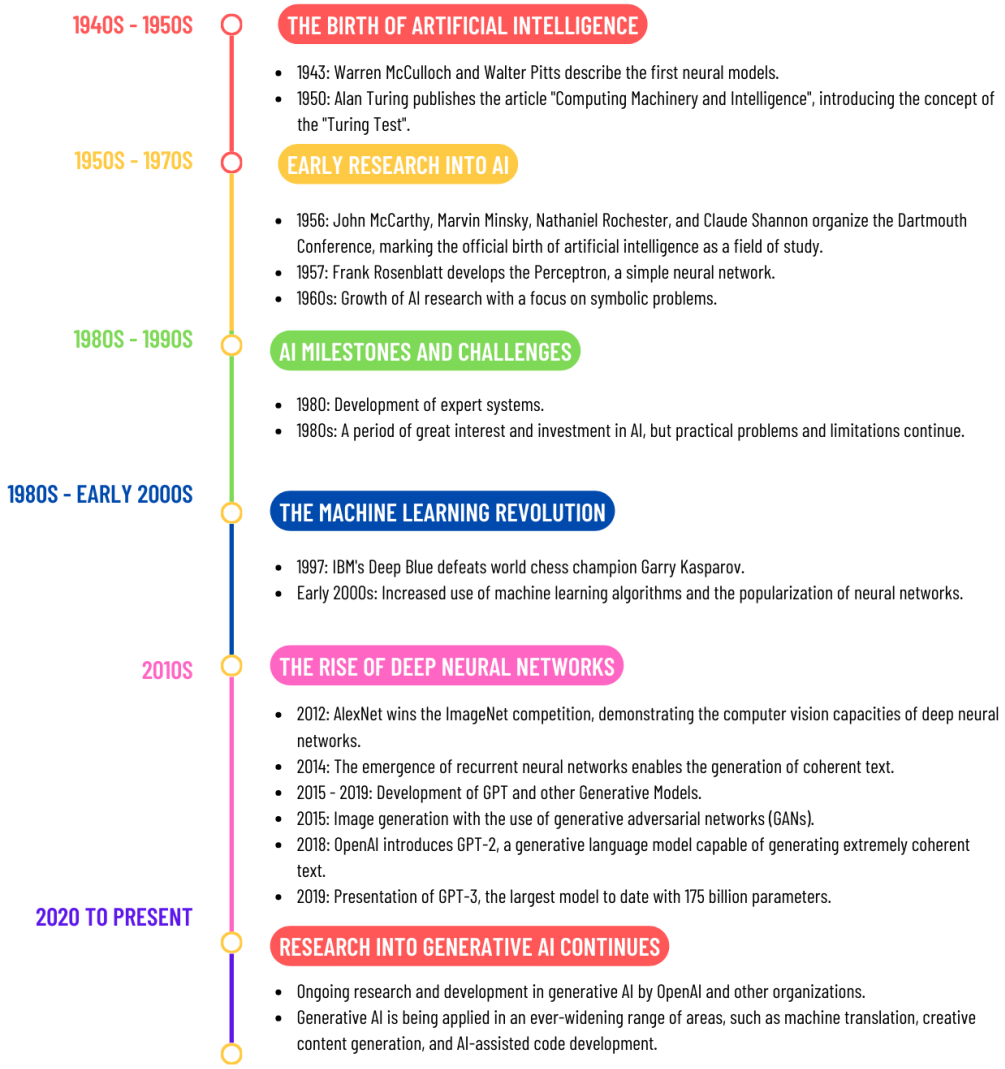
Generative artificial intelligence (GAI) is a branch of artificial intelligence (AI) based on the development of data models that are capable of generating new content. Unlike other AI methods that are limited to analysing existing data or solving specific problems, GAI is capable of imagining, creating, and producing information, images, text and other forms of content.

GAI relies on neural networks, which can learn complex patterns and features from training datasets, using the learned information to generate new outputs that resemble the original data and are consistent with it.

The GAI tool GPT-3.5 (generative pre-trained transformer 3.5) developed by the company OpenAI took the world by storm in late 2022. These models are able to understand and generate natural language in surprisingly coherent and diverse ways.

But it wasn't the first AI to be created, nor is it the only type of AI out there.

The AI story



Source: author - *Link to the timeline*

It is necessary to classify the different types of AI featured in this timeline: machine learning is not the same thing as deep learning, for example.

How do we classify AI?

Artificial intelligence can be classified in several ways, such as by how it learns or functions, its features or its aims. We will focus on classification by type of learning.

- **Machine learning**. Computer systems are set up to use data to learn and improve, without requiring explicit programming. There are three main subcategories:
    - **Supervised learning**. Labelled datasets are used to train algorithms that can then make accurate predictions about new data. It is used, for example, in applications such as email sorting (to differentiate spam from non-spam) and medical diagnoses (e.g. diagnosing melanomas).
    - **Unsupervised learning**. The models learn from unlabelled data to discover hidden patterns or to cluster data. This includes tasks such as data clustering and dimensionality reduction, which are useful for data analysis and customer segmentation, for example.
    - **Reinforcement learning**. The models learn through interacting with a dynamic environment and make decisions with feedback in the form of rewards or penalties. It is used, for example, in applications such as autonomous driving and training robots.
- **Deep learning**. A branch of artificial intelligence that uses neural network models to learn and understand complex patterns in data. Its features include:
    - Deep neural networks. They consist of multiple layers of interconnected neurons inspired by how the human brain works. Each layer processes information and transmits the output to the next layers, allowing hierarchical learning.
    - Representational capacity. With multiple layers, this type of AI can learn to represent highly complex data patterns, making it suitable for tasks such as image recognition, natural language processing, and other areas involving complex data.
    - Hierarchical learning. Successive layers learn representations of data with multiple levels of abstraction. For example, in image recognition, the initial layers detect basic shapes, while deeper layers identify more complex patterns.
    - Input processing with many dimensions. Dimensions are structures identifying the different attributes or variables that describe each instance of the data. The ability to manipulate large amounts of data with many dimensions is key to many applications, as it allows models to capture subtle and complex details in the data.
    - Large quantities of data and computing power. Huge quantities of data are required to train these models in order to obtain the best results, calling for enormous computing power.
    - The applications of deep learning AI.
        - Image recognition. Identifying objects and patterns in images, e.g. distinguishing cats from dogs.
        - Natural language processing (NLP), e.g., machine translation, sentiment analysis and text generation.

- Computer vision. Facial recognition, object segmentation, etc.
- Voice recognition. Converting voice to text and vice versa.
- Computer games. Learning optimal policies in games such as Go and video games.
- Programming language code. Generating or modifying program code.

Generative artificial intelligence falls within the deep learning category of AI. A graphic representation would look like this:



Source: author

This form of artificial intelligence, GAI, learns automatically from data. It analyses immense amounts of data, finds patterns and relationships that would otherwise be difficult to detect and creates original material from that data.

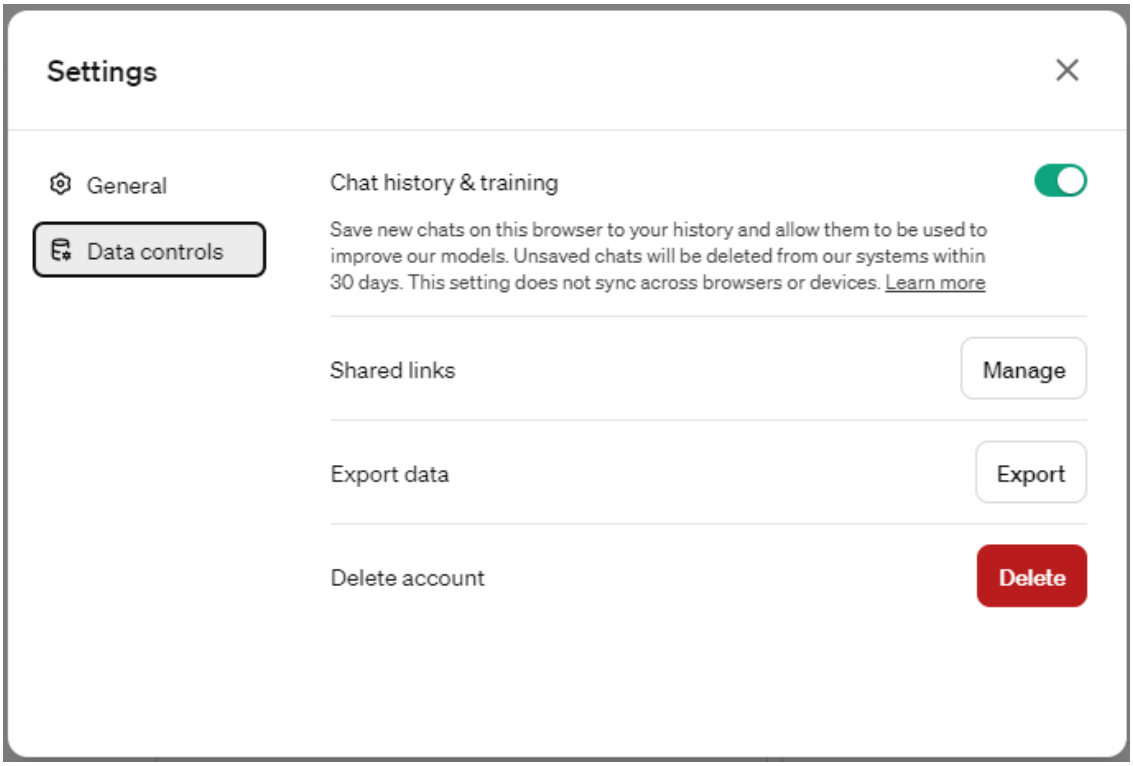
Generative AI uses generative adversarial networks (GANs) that are based on two neural networks, a *generator* and a *discriminator*. The generator is responsible for creating new content, while the discriminator differentiates between genuine and simulated data. This interaction between generator and discriminator allows AI to learn from data effectively and efficiently, generating high-quality, realistic new content.

How are GAIs fed?

We don't know what all the sources used to train the different GAI models are, but those we do know include:

- Wikipedia: the collaborative encyclopaedia.
- *The New York Times*: a US-based international newspaper.
- Reddit: one of the largest social media platforms, dedicated to discussions of different topics.
- arXiv: a repository of scientific articles.
- GitHub: a code hosting platform.
- OpenSubtitles: a platform hosting film and TV series subtitles.
- Etc.

However, it should also be noted that many GAI applications use the input we provide to ask questions as a source of information for their models. Some GAI applications allow us to request that the data entered do not feed the model. For example, in ChatGPT, we can disable this option in the settings section:



Source: author. ChatGPT settings

What are the benefits of GAI?

- **Creative content generation:**
  - GAI is used in the creation of generative art, product design and other fields that need original content to be produced.
  - Creative content such as images, text, video and music can be produced.
- **More data for training:**
  - For machine learning tasks, GAI can be used to augment training datasets, generating new samples that enrich the variety and complexity of the data available to the model.
- **Scenario simulation:**
  - GAI can be used to simulate situations and environments to train AI models, such as those based on reinforcement learning, where interactions with the environment need to be simulated.
- **Style and learning transfer:**
  - Style transfer can be used to apply visual styles or specific features from one data class to another. For example, it can be asked to write a poem in the style of a particular poet.
  - It allows learning to be transferred between domains, such as applying the style of a work of art to a photograph.

- **Custom content creation:**
    - GAI can generate personalized content based on user preferences. For example, generating personalized product recommendations.
- **Development of dialogue and natural language systems:**
    - In the field of natural language processing, GAI helps create more advanced dialogue systems and generate coherent and natural text.
- **Medical simulations:**
    - GAI can be used to simulate medical data to train models in the field of medicine and health.
- **Automatic design:**
    - GAI is used in product design, such as the automatic generation of architectural, graphic and other designs.
- **Artificial intelligence research:**
    - GAI is used as a tool for research in other areas of AI, such as computer vision, natural language processing, and artificial intelligence in general.
- **Numerical data generators:**
    - GAI can generate numbers or numerical data that follow a certain pattern or distribution.
- **Virtual reality (VR) and augmented reality (AR) generators:**
    - These can create virtual scenarios, objects, and interactions in a virtual or augmented reality environment.
- **3D modelling generators:**
    - 3D objects or scenarios can be generated for a range of applications, such as video game design or simulation.

In brief, generative artificial intelligence has enormous potential to create new and unique content, and it has applications in a wide range of sectors including art and design, simulating environments and improving AI models.

GAI tools

1 TEXT TO TEXT

- **Input**: text.
- **Output**: text.
- **Applications**: translating, information searches, text summaries, keywords, text correction, etc.
- **Tools**: ChatGPT, Bing Chat, Perplexity, Tutor AI, Google Bard AI, Writesonic, Canva, ChatPDF, Consensus App, Chat Doc, Humata AI, Explain Paper, Peer, Magical School, Quizz, Learnt AI, Talk to Books and Harpa AI.

2 TEXT TO IMAGE

- **Input**: text.
- **Output**: image.
- **Applications**: generating images, seeking inspiration, creating art, avatars and logos.
- **Tools**: Dall-E, Midjourney, Bing Image Creator, Stable Diffusion, Adobe Firefly, Google Bard AI, Scribble Diffusion, Craiyon and Canva.

3 TEXT TO AUDIO

- **Input**: text.
- **Output**: audio.
- **Applications**: generating audio files, audiobooks and podcasts; creating voiceovers with other people's voices or your own; translating videos, creating melodies with selected parameters and modifying other passages of music.
- **Tools**: Play HT, Murf AI, Listnr, Speechify, Resemble AI, Podcast AI, WellSaid and ElielevenLabs.

4 TEXT TO VIDEO

- **Input**: text.
- **Output**: video.
- **Applications**: generating videos with selected specific features, editing and translating videos.
- **Tools**: Movio, Lumen5, Synthesia.io, D-ID, Runway, Fliki and Pictory.

5 IMAGE TO IMAGE

- **Input**: image.
- **Output**: image.
- **Applications**: modifying images (making objects or people disappear, changing objects, removing backgrounds, etc.).
- **Tools**: Canva, Playground AI, Fotor, Autoenhance AI and Hotpot AI.

6 TEXT TO CODE

- **Input**: text.
- **Output**: code.
- **Applications**: generating, improving and documenting code.
- **Tools**: ChatGPT, GitHub Copilot, SourceAI, Replit Generate Code and Fliplet.

7 TEXT TO PRESENTATIONS

- **Input**: text.
- **Output**: presentations.
- **Applications**: generating slides.
- **Tools**: Tome, ClassPoint.io, Gamma App, GPT for Slides, Beautiful AI and Motionit AI.

8 AUDIO TO TEXT

- **Input**: audio.
- **Output**: text.
- **Applications**: transcribing audio into different languages.
- **Tools**: Dictation.io, Happy Scribe and Whisper.

9 VIDEO TO TEXT

- **Input**: video.
- **Output**: text.
- **Applications**: transcribing videos, transcribing and translating.
- **Tools**: Happy Scribe, Saylient.io, Rask AI and Notta AI.

How to interact with AI

Introduction

We interact with generative artificial intelligence by giving a language model (ChatGPT, DALL-E, etc.) a short instruction or description to guide the responses it generates. This instruction or description is called a **prompt**.

By providing a specific prompt, we can influence the type of responses generated by the model. The choice of prompt is decisive, as it directly affects the nature and quality of the response generated. Experimenting with different prompts can help you obtain more accurate and useful results for the task you want to carry out.

Example of the procedure used by a GAI model to respond to a text prompt

Once the GAI model has been trained, generating a text response to a written prompt involves the following steps:

- The prompt is divided into smaller units (called tokens).
- Statistical patterns are used to predict likely words or phrases that could form a coherent answer to the question.
- Patterns of words and phrases found in the big data model (stored in neural networks) are identified.
- GAI uses these patterns to calculate the probability that specific words or phrases will appear in a given context and generate predictions.
- Starting with a random prediction, GAI uses these probabilities to predict the next likely word or phrase in its response.
- Predicted words or phrases become readable text.
- The readable text is filtered to remove any offensive content.
- These final steps are repeated until the response is complete.
- The answer is reprocessed to improve it (format, punctuation, etc.).

How do you formulate prompts to improve the results obtained with GAI?

Specifying the context:

- Provide enough context for the GAI model to understand the task.
- Include relevant details that help guide the response.

Clear instructions:

- Provide clear and direct instructions.
- Avoid ambiguity in the prompt.

Search target:

- Clearly identify the target of the search: what information are you looking for?
- Identify what type of response you expect to receive (descriptive information, instructional information, creative information, etc.).

Keywords:

- Include keywords that are fundamental to the task. These help guide the result.

- Use specific terms that you want the answer to contain.

Constraints and conditions:

- Set clear constraints (if necessary).
- Include specific conditions that the GAI must meet.

Prompt length:

- Check that the length of the prompt is appropriate for the complexity of the task.
- Avoid prompts that are too long or too short.

Specific examples:

- Provide specific examples to illustrate what you are looking for.
- Keep in mind that the examples will help the GAI model to understand the task better.

Direct questions:

- Ask direct questions if you are looking for specific answers.
- Formulate questions clearly.

Iterative feedback.

- Decide whether you need to include instructions on how to adjust the response if it doesn't meet your expectations.
- Provide examples of how to improve or change the output provided by the GAI model.

Vocabulary and style:

- Indicate the type of vocabulary you want to see in the response or the desired style.
- Specify the particular tone that the tool should adopt.
- Indicate if you want a formal, informal, creative or other response.

Review and adjustment:

- Check the prompt before sending it to ensure there are no errors or misunderstandings.
- Make adjustments and experiment with different reformulations of the prompt.

Jordi Linares, in his article *The Art of prompting: Empowering students to harness the power of LLMs (ChatGPT and others) in education*, explains another way to generate good results by using more than one prompt. This involves taking a **top-down approach**: starting by asking for a global view of the topic then progressively diving deeper to get more detail.

Some image-generating GAI models require prompts to use specific syntax. For example, Image Creator (https://www.bing.com/create) requires the following pattern:

adjective + noun + verb + style

For example: futuristic, urban landscape, reflecting sunset, conceptual art

Examples of prompts

Prompt: Create a story for a six-year-old boy to help him overcome shyness.

Output:

ChatGPT

Rodolfo's Treasure

In a small village, Rodolfo, a curious little mouse, discovered a golden coin. Excited to show it to his friends, he felt shy when he saw them busy playing. Determined, he juggled the coin, creating a small show that surprised and delighted everyone. Rodolfo learned that sharing what we love can overcome embarrassment, realizing that each one has something special to offer. Since then, he and his friends lived happily, celebrating the unique wonders each brought to the group.

Prompt: Create a story for six-year-old children in the style of René Goscinny, creator of Asterix and Obelix. The hero's name is Pere, a six-year-old boy with blue eyes and blond hair. The style of the story should be adventurous and exciting, based on the fantasy genre. In the story Pere learns how to manage emotions, as he is very shy. Some of the characters to include are his father, Joan; mother, Marta; his sister, Èlia, and Ignasi, his best friend. Pere enjoys reading and athletics. Start the story with an entertaining title and divide it so each part starts on a new page, leaving a space to add images.

Output:

ChatGPT

Title: Pere's Daring Quest

Page 1: The Mysterious Forest Discovery

In a quaint village, Pere, a six-year-old boy with blue eyes and blond hair, discovers a hidden passage in the mysterious forest. Guided by his insatiable curiosity, he stumbles upon an ancient scroll that hints at the magical Garden of Emotions. Intrigued, Pere decides to embark on a quest to find this enchanted place that promises to help him overcome his shyness.

Page 2: The Enchanted Garden Adventure

Accompanied by his best friend, Ignasi, Pere follows the clues from the ancient scroll, leading them to the Enchanted Garden. Along the way, they encounter talking animals, friendly fairies, and tricky puzzles that challenge Pere to face his fears. The magical beings in the garden teach Pere the importance of embracing his emotions and finding strength in vulnerability.

Page 3: Pere's Triumph and Heartfelt Lessons

As Pere and Ignasi exit the Enchanted Garden, they carry with them a magical flower that blooms with courage. Back home, Pere shares his incredible adventure with his family—his father, Joan; his mother, Marta; and his sister, Èlia. Inspired by Pere's bravery, his family celebrates with a special dinner. Pere learns that it's okay to feel shy sometimes, but facing those feelings can lead to incredible discoveries. The story ends with Pere and his family enjoying a starlit night, knowing that the most extraordinary adventures often start with a curious heart.

Prompt: A ten-year-old boy with brown eyes and light brown hair. Sitting on the floor gazing at the stars through the window. In a watercolour style.

Output:



Prompt: A 25-year-old Catalan woman, in a yellow dress, studying at her desk with a blue picture stamped with the letters "UOC". Foreground, natural light, cool wall colours, production quality, depth of field, professional and realistic photography, lots of detail, long exposure time, f/1.8, diffuse backlighting, studio lighting.

Output:



Prompt: Please find and indicate any errors in the following code written in the PASCAL programming language, whose purpose is to count the number of lowercase or uppercase "a" characters in a sentence ending in a full stop entered by the user.

```
Program lettera;
Uses crt;
var cont:integer;
letter:char;
Begin clrscr;
writeln('As counter. Write a line ending in a full stop');
writeln('_____');
cont:=0;
```

```
read(letter);
while (letter<>'.') ;
begin if ((letter='a') or (letter='A')) then begin cont:=cont+1;
end; read(letter);
end;
write('Amount of As: ',cont); readkey;
end.
```

Output:

ChatGPT

Here's the corrected version of your Pascal code:

pascal — Copy code

```
Program lettera;
Uses crt;
var
  cont: integer;
  letter: char;

Begin
  clrscr;
  writeln('As counter. Write a line ending in a full stop');
  writeln('_____________________________________________');
  cont := 0;

  repeat
    read(letter);
    if (UpCase(letter) = 'A') then
    begin
      cont := cont + 1;
    end;
  until (letter = '.');

  writeln('Amount of As: ', cont);
  readkey;
End.
```

Changes made:

1. Replaced the curly single quotes (`‘’`) with straight single quotes (`''`) for consistency.
2. Used the `UpCase` function to convert the input character to uppercase before comparing it with 'A'. This ensures that both uppercase and lowercase 'a' characters are counted.
3. Replaced the `while` loop with a `repeat...until` loop to correctly read characters until a full stop is encountered.

The program should now correctly count the number of "a" characters (lowercase or uppercase) before the full stop marking the end of the entry.

Additional information can be found here:

Gómez Cardosa, D. and Garcia Brustenga, G. (2023). *How to question AI. Useful prompts for teaching staff when using generative AI*. https://openaccess.uoc.edu/bitstream/10609/147885/3/Manual%20IA_ENG_PROMTS.pdf

Ethical criteria for the use of AI

Introduction

Generative AI raises a range of issues that need to be addressed not just by the entities that develop it but by all of society, users and non-users:

1. **Ethics and responsibility**. Generative AI can create content that is misleading, offensive, or even illegal. This raises ethical questions about who is responsible for the content generated and how control algorithms can be implemented to prevent abuse.
2. **Biased training data**. Generative AI models learn from data, and if the data used to train them are biased (cultural biases, gender, etc.), AI can learn from these biases and replicate them in the content it generates.
3. **Controlling the information generated**. It is difficult to ensure full control over the content generated by GAI. This may lead to situations where the generated content does not reflect the user's intentions or may be misinterpreted.
4. **Disinformation and manipulation (deepfakes)**. The ability to generate convincing content can be used to create fake news or manipulate information.
5. **Copyright and intellectual property**. Who owns the rights to AI-generated content? This is a complex legal issue that has not yet been clearly settled.
6. **Data privacy and security**. Generative AI models store and process large amounts of data. It is important to ensure that these data are protected and not used improperly. Are there concerns about the misuse of personal information?
7. **Generation of harmful content**. AI can be used to generate content that is offensive, discriminatory, or harmful. How should this type of content be handled?
8. **Algorithm transparency.** The methods used to train and operate the algorithms in generative AI can be opaque and difficult for people outside the field to understand.
9. **Equitable adoption and access**. How do we ensure that the benefits of generative AI are distributed equitably and do not widen social and economic gaps?
10. **Regulations and standards**. The laws and regulations governing the use of AI are still evolving and may be difficult to enforce in this innovative environment. In fact, in many cases they are just recommendations rather than regulations.
11. **Energy consumption and carbon footprint**. Training and using large generative AI models can require immense amounts of energy. This is problematic if this energy comes from non-renewable sources that generate greenhouse gas emissions.
12. **Impact on the labour market**. Could the adoption of generative AI technologies have implications for jobs in some sectors? Will new jobs be generated in the same proportion as the jobs that are destroyed?
13. **Authorship**. How do we ensure people can distinguish between human-created and generative AI-generated content? This is essential for transparency and for people to trust the information they use.
14. **Inequality of access and concentration of power**. Access to AI technology may not be equitable, and companies or organizations with more resources may have an unfair advantage when using these tools. This could lead to power becoming more concentrated and greater social inequality.

Addressing these problems and challenges will require a combination of ethical conduct, regulatory oversight and ongoing technological development. It is important that communities around the world, including researchers, companies, policymakers and multidisciplinary teams from different branches of science (philosophers, physicists, mathematicians, computer scientists, linguists, sociologists, etc.) work together to find sustainable and responsible solutions.

There is currently **no regulatory supervision** of the use of GAI. There are just **recommendations** on its use.

The 2019 Beijing Consensus on Artificial Intelligence and Education contains guidelines on how to apply a humanistic approach to AI in education.

UNESCO has published a range of recommendations and guidelines on the ethical, safe and equitable use of generative AI in education and research.

UNESCO's November 2021 Recommendation on the ethics of artificial intelligence provides a regulatory framework for addressing the controversies surrounding artificial intelligence, including in education and research.

In 2021 UNESCO also published *AI and education: guidance for policy-makers*, setting out specific recommendations for developing policies on the use of AI in education.

In 2023 UNESCO issued the following document of specific recommendations: *Guidance for generative AI in education and research*.

UNESCO's review of current national AI strategies shows that countries are adopting different policy responses, ranging from banning GAI to assessing the need to adapt existing frameworks or urgently formulating new regulations.

In short, the regulatory supervision of GAI in education and other areas requires a series of policy steps and measures based on a human-centred approach to ensure it is used ethically, safely and equitably.

Considerations for UOC students

Appropriating someone else's idea and presenting it as your own or copying and pasting a text without citing the source are considered to be plagiarism and the Universitat Oberta de Catalunya (UOC) treats this as a serious offence.

Source: UOC. (2023). Academic plagiarism

https://biblioteca.uoc.edu/en/biblioguides/biblioguide/Academic-plagiarism/

To avoid committing this offence, you must **cite text** produced by others. This UOC web page gives examples of how texts can be cited depending on the source and citation style used.

Using the APA style guide, information obtained from a GAI source should be cited as follows:

Organization responsible for the AI tool. (Year of AI query). *Tool name in italics* (tool version) [additional description]. URL

For example:

OpenAI. (2023). *ChatGPT* (January 2022 version) [*prompt:* "**What is the UOC?**"].

https://chat.openai.com

If the prompt used was very long, it is recommended that the citations be listed in an appendix to the paper, with references.

Considerations for university students on the use of GAI in their work at the UOC:

- UOC students make a commitment to meet the university's standards. The UOC calls on you to use GAI responsibly in line with your ethics, academic integrity and concern for your own reputation.
- Not all UOC courses allow the use of GAI, so you should ask teaching staff for permission to use these tools to complete part of your course assignments.

- All tasks done using GAI must be cited. In the absence of a citation, teaching staff may request additional information to verify authorship, or even set an alternative test or oral examination. If you do not do the test or cannot prove your authorship, this could be treated in the same way as plagiarism, which is serious misconduct, and you may fail the course.
- Students are ultimately responsible for the content they submit in the assessment activities. All information obtained using GAI tools must, therefore, be verified.
- If the GAI tool has a setting allowing the outputs generated to be fed back into it, it should be disabled.

Most common uses of AI in learning

Some specific GAI applications that could benefit students in their learning process include:

- **Research and writing**:
    - Content generation. Students carrying out research can use GAI models to generate paragraphs related to their subject from keywords or main ideas.
    - Writing suggestions. The tools could offer specific suggestions to improve the consistency, style and grammar of a piece of writing, providing an instant revision.
- **Study aids:**
    - Problem-solving skills. Students struggling to understand a mathematical or scientific problem can use a generative model to obtain a step-by-step explanation.
    - Model generation. AI can generate visual models for complex mathematical or scientific concepts, making them easier for students to understand.
- **Day-to-day work:**
    - GAI can help with the tasks that students regularly have to perform in their academic lives: translating, correcting or summarizing texts; searching for information; searching for ideas to complete assignments; generating images; improving presentations; generating audio files; translating, transcribing or generating videos; improving code written in a programming language, etc.
- **Learning new skills:**
    - Personalized tutorials Students who want to learn a new skill, such as programming, could receive tutorials generated specifically for their level of experience.
    - Interactive simulations. GAI can create interactive simulations for the practical learning of complex concepts in certain disciplines, such as physics or biology.
- **Exam preparation:**
    - Generation of exercises and questions. When revising a subject, students could generate sets of exercises and questions covering specific areas they want to review.
    - Review of key concepts. GAI can generate concise summaries and reviews of key concepts for quick revision.
- These tools can be extremely useful. Nevertheless, it is absolutely vital that students use them sensibly, verify the authenticity of the content generated and adhere to the ethical and academic standards set by the UOC.

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